

Business Intelligence

Practice Problems Work-Through

① A simple invoice-screening perceptron uses one neuron.

a) Given

$$x = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad w = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \quad b = 1$$

Compute input

$$z = w^T x + b$$

$$\begin{aligned} \Rightarrow w^T x &= \begin{bmatrix} 1 & 0 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = 1 \cdot 2 + 0 \cdot (-1) + (-2) \cdot (3) \\ &= 2 - 6 = -4 \end{aligned}$$

$$w^T x + b = -4 + 1 = -3$$

$$\Rightarrow z = -3$$

b) Use the decision rule

$$\hat{y} = \begin{cases} 1, & z \geq 0, \\ 0, & z < 0. \end{cases}$$

Determine the predicted class.

$$\text{Prediction: } \hat{y} = 0$$

$\Rightarrow$ invoice rejected.

c) Repeat the same calculation for

$$x' = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

Using the same weights w and bias b .

$$\begin{aligned} \Rightarrow w^T x' &= \begin{bmatrix} 1 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = 1 \cdot 1 + 0 \cdot 2 + (-2) \cdot 0 \\ &= 1 \end{aligned}$$

$$\Rightarrow z = w^T x' + b = 1 + 1 = 2$$

Prediction: $\hat{y} = 1$ (invoice accepted)

③ A credit pre-check model uses one hidden layer and one output neuron.

a)

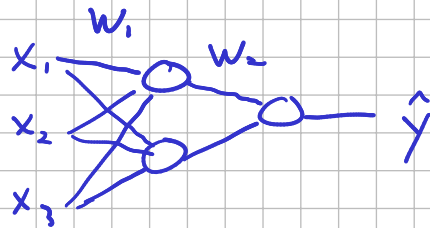
$$x = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad W_1 = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 2 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Compute :

$$z^{(1)} = W_1 x + b_1$$

$$a^{(1)} = \text{ReLU}(z^{(1)}).$$

$\Rightarrow$ Let's start by considering the architecture. We know that the network has 3 inputs, 2 hidden neurons (see shape W_1), and one output.



$$\Rightarrow z^{(1)} = W_1 x + b_1 :$$

$$W_1 x = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 2 + 1(-1) + 0(1) \\ -1(2) + 0(-1) + 2(1) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$z^{(1)} = W_1 x + b_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$a^{(1)} = \text{ReLU}(z^{(1)}) = \begin{bmatrix} \text{ReLU}(1) \\ \text{ReLU}(1) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

* Where $\text{ReLU}(z) = \max(0, z)$

b) Given

$$W_2 = [2 \ -1], \quad b_2 = -1$$

Compute the output score

$$z^{(2)} = W_2 a^{(1)} + b_2.$$

$\Rightarrow$ This is the exact same calculation as in the first layer, only we use the activation output from layer 1 as our input.

$$\Rightarrow W_2 a^{(1)} = [2 \ -1] \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2 \cdot 1 + (-1)(1) = 1$$

$$z^{(2)} = W_2 a^{(1)} + b_2 = 1 + (-1) = 0$$

$$\boxed{z^{(2)} = 0}$$

c) Use the decision rule

$$\hat{y} = \begin{cases} 1, & z^{(2)} \geq 0, \\ 0, & z^{(2)} < 0. \end{cases}$$

$\Rightarrow$ Since $z^{(2)} = 0$, by the given decision rule, our prediction is:

$$\boxed{\hat{y} = 1}$$

$\Rightarrow$ The pre-credit check for the applicant has passed!

⑧ A single linear neuron.

a) Given

$$x = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad b = 0, \quad y = 3$$

Compute the forward pass

$$\hat{y} = w^T x + b$$

and the loss

$$L = \frac{1}{2} (\hat{y} - y)^2$$

$$\Rightarrow \hat{y} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 0 = \boxed{2} \checkmark$$

$$L = \frac{1}{2} (2 - 3)^2 = \boxed{\frac{1}{2}} \checkmark$$

b) Compute the gradients $\frac{\partial L}{\partial w}$, $\frac{\partial L}{\partial b}$.

$\Rightarrow$ We re-write our loss function to express in terms of w and b

$$L = \frac{1}{2} (\hat{y} - y)^2 = \frac{1}{2} [(w^T x + b) - y]^2$$

$$\Rightarrow \text{Chain rule:} \quad \begin{aligned} h(x) &= f \circ g(x) = f(g(x)) \\ h'(x) &= f'(g(x)) g'(x) \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial w} &= 2 \left(\frac{1}{2} \right) (w^T x + b - y) x \\ &= (2 - 3) x = -x = - \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \end{aligned}$$

$$\frac{\partial L}{\partial b} = 2 \left(\frac{1}{2} \right) (w^T x + b - y) (1) = (2 - 3) (1) = -1$$

c) Using the learning rate $\eta = 0.1$, compute updated parameters w_{new} and b_{new}

$\Rightarrow$ Use the update rule from Prof. H4:

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla_{\theta} L$$

$$\frac{\partial L}{\partial w} = \nabla_w L$$

$$\Rightarrow w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w}$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} -0.2 \\ 0.1 \end{bmatrix}$$

$$w_{\text{new}} = \begin{bmatrix} 1.2 \\ -0.1 \end{bmatrix}$$

$$\Rightarrow b_{\text{new}} = b_{\text{old}} - \eta \frac{\partial L}{\partial b}$$

$$= 0 - 0.1(-1)$$

$$b_{\text{new}} = 0 + 0.1 = 0.1$$

⑫ An upsell classifier uses a hidden ReLU layer and a sigmoid output.

a) Given

$$x = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad w_1 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$w_2 = [1 \ -1], \quad b_2 = 0, \quad y = 1,$$

compute the forward pass

$$a^{(1)} = \text{ReLU}(z^{(1)}), \quad \hat{y} = \sigma(z^{(2)}).$$

$\Rightarrow$ We see from the shapes of the inputs, weights, and output, that the architecture is structurally identical to that in problem ③. The output here, however, is a sigmoid activation function.

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$\Rightarrow$ Approach: go layer by layer!

Layer 1

$$\Rightarrow z^{(1)} = w_1 x + b_1$$

$$\Rightarrow w_1 x = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$
$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1(1) + (-1)(0) + 0(1) \\ 0(1) + (1)(0) + 1(1) \end{bmatrix}$$
$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow z^{(1)} = w_1 x + b_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = z^{(1)}$$

$$\Rightarrow a^{(1)} = \text{ReLU}(z^{(1)}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Layer 2

$$\Rightarrow z^{(2)} = W_2 a^{(1)} + b_2$$

$$\Rightarrow W_2 a^{(1)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1(1) + (-1)(1) = 1 - 1 = 0$$

$$z^{(2)} = W_2 a^{(1)} + b_2 = 0 + 0 = 0 = z^{(2)}$$

$$\Rightarrow \hat{y} = \sigma(z^{(2)})$$

$$= \frac{1}{1 + e^{-z^{(2)}}} = \frac{1}{1 + e^0}$$

$$\hat{y} = \frac{1}{2}$$

b) Using binary cross-entropy, compute the error signals:

$$\delta^{(2)} = \hat{y} - y, \quad \delta^{(1)}$$

$$\Rightarrow \delta^{(2)} = \hat{y} - y = \frac{1}{2} - 1 = -\frac{1}{2}$$

$$\delta^{(2)} = -0.5$$

$\Rightarrow$ To find $\delta^{(1)}$, we have to carry $\delta^{(2)}$ back through the network.

$\Rightarrow$ To do this, we have to pull it back through the weights W_2 , and gate it with the derivative of the hidden layer activation function.

ReLU



$\Rightarrow$ derivative is the slope!
(1)

$$\Rightarrow \delta^{(1)} = \underline{(W_2^T \delta^{(2)})} \odot \text{ReLU}'(z^{(1)})$$

$$\Rightarrow W_2^T \delta^{(2)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} (\sim 0.5) = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix}$$

$\Rightarrow \text{ReLU}'(z^{(1)}) \rightarrow$ Since $z^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and both elements are positive, then the derivative of the ReLU function is 1 for each

$$\text{ReLU}'(z^{(1)}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

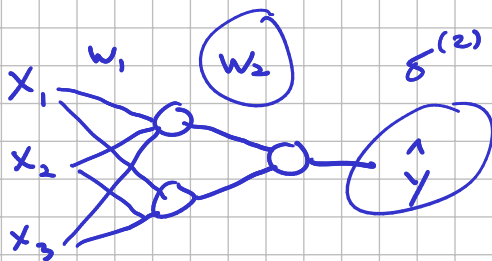
$$\Rightarrow (W_2^T \delta^{(2)}) \odot \text{ReLU}'(z^{(1)}) = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix} \odot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix}$$

↑
element-wise multiplication

c) Compute all gradients and update W_2, b_2, W_1, b_1

General rule for gradients: Error $\times$ Input (from left)

Example:



$$\nabla_{W_2} L = \delta^{(2)} (a^{(1)})^T$$

⇒ Work backwards from the output!

$$\nabla_{W_2} L = \delta^{(2)} (a^{(1)})^T$$

$$a^{(1)} =$$

$$= -0.5 [1 \ 1] = [-0.5 \ -0.5]$$

$$\nabla_{b_2} L = \delta^{(2)} = -0.5$$

⇒ From Prof. H4's lecture 26.03.2026:

"The error becomes the bias of the previous layer..."

⇒ This is a good clue!

Update $\eta = 0.2$

$$\begin{aligned} W_{2(\text{new})} &= W_{2(\text{old})} - \eta \nabla_{W_2} L \\ &= [1 \ -1] - 0.2 [-0.5 \ -0.5] \\ &= [1 \ -1] - [-0.1 \ -0.1] \end{aligned}$$

$$W_{2(\text{new})} = [1.1 \ -0.9]$$

$$\begin{aligned} \Rightarrow b_{2(\text{new})} &= b_{2(\text{old})} - \eta \nabla_{b_2} L \\ &= 0 - 0.2 (-0.5) \\ &= 0 - (-0.1) \end{aligned}$$

$$b_{2(\text{new})} = 0.1$$

⇒ Now we do the hidden layer. We use

$$\delta^{(1)} = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix} \text{ and } x = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Gradients:

$$\nabla_{W_1} L = \delta^{(1)} x^T$$

$$= \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.5 & 0.0 & -0.5 \\ 0.5 & 0.0 & 0.5 \end{bmatrix}$$

$$\nabla_{b_1} L = \delta^{(1)} = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix}$$

Updates:

$$W_{1(\text{new})} = W_{1(\text{old})} - \eta \nabla_{W_1} L$$

$$= \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} - 0.2 \begin{bmatrix} -0.5 & 0.0 & -0.5 \\ 0.5 & 0.0 & 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} - \begin{bmatrix} -0.1 & 0.0 & -0.1 \\ 0.1 & 0.0 & 0.1 \end{bmatrix}$$

$$W_{1(\text{new})} = \begin{bmatrix} 1.1 & -1.0 & 0.1 \\ 0.1 & 1.0 & 0.9 \end{bmatrix}$$

$$\Rightarrow b_{1(\text{new})} = b_{1(\text{old})} - \eta \nabla_{b_1} L$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.2 \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -0.1 \\ 0.1 \end{bmatrix}$$

⇒

$$b_{1(\text{new})} = \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}$$

(20) A campaign response model uses 4 inputs, 2 hidden ReLU layers, and a 3-class softmax output.

a) Given

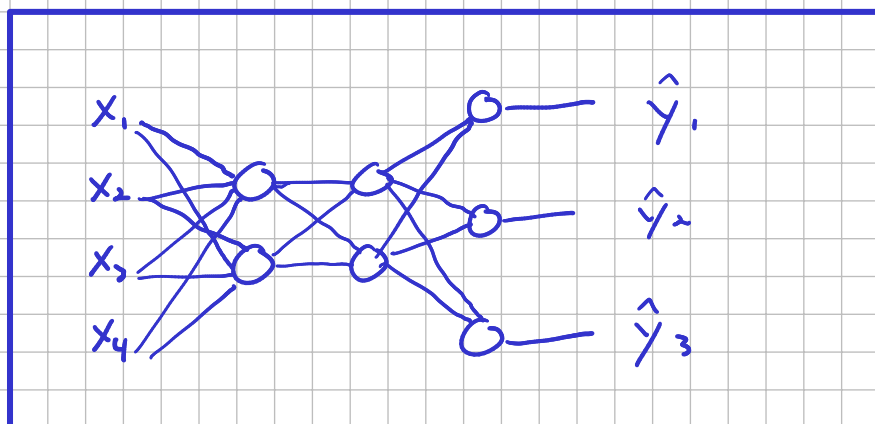
$$x = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}, \quad W_1 = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} -2 \\ 0 \end{bmatrix},$$

$$W_3 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 0 \end{bmatrix}, \quad b_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad y = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

compute the forward pass

$$z^{(1)}, a^{(1)}, z^{(2)}, a^{(2)}, z^{(3)}, a^{(3)}, \hat{y}$$



* Note that the number of neurons in each layer matches the number of rows in its weight matrix!

$\Rightarrow$ Go layer by layer!

Layer 1

$$z^{(1)} = W_1 x + b_1$$

$$W_1 x = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 0 + 0 \cdot (-1) + 1 \cdot 2 \\ 0 \cdot 1 + 1 \cdot 0 + (-1) \cdot (-1) + 2 \cdot 0 \end{bmatrix}$$
$$= \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$z^{(1)} = W_1 x + b_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \boxed{\begin{bmatrix} 3 \\ 1 \end{bmatrix}} = z^{(1)}$$

$$\boxed{a^{(1)} = \text{ReLU}(z^{(1)}) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}}$$

Layer 2

$$z^{(2)} = W_2 a^{(1)} + b_2$$

$$\Rightarrow W_2 a^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$z^{(2)} = W_2 a^{(1)} + b_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

$$\boxed{z^{(2)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

$$\boxed{a^{(2)} = \text{ReLU}(z^{(2)}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

Layer 3

$$z^{(3)} = W_3 a^{(2)} + b_3$$

$$W_3 a^{(2)} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$z^{(3)} = W_3 a^{(2)} + b_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$z^{(3)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum e^{z_j}}$$

$\Rightarrow$

$$z_1^{(3)} = 0$$

$$z_2^{(3)} = 0$$

$$z_3^{(3)} = 0$$

$$\Rightarrow \begin{bmatrix} e^0 / 3e^0 \\ e^0 / 3e^0 \\ e^0 / 3e^0 \end{bmatrix} =$$

$$\begin{bmatrix} 0.333 \\ 0.333 \\ 0.333 \end{bmatrix} = \hat{y}$$

b) Use cross-entropy with SoftMax to compute

$$\delta^{(3)} = \hat{y} - y, \delta^{(2)}, \delta^{(1)}$$

$$\delta^{(3)} = \begin{bmatrix} 0.333 \\ 0.333 \\ 0.333 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.333 \\ -0.667 \\ 0.333 \end{bmatrix}$$

$\Rightarrow$ Propagate $\delta^{(3)}$ back to get our $\delta^{(2)}$:

$$\Rightarrow \delta^{(2)} = (W_3^T \delta^{(3)}) \odot \text{ReLU}'(z^{(2)})$$

$$W_3^T \delta^{(3)} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.333 \\ -0.667 \\ 0.333 \end{bmatrix} = \begin{bmatrix} 0.333 + 0.667 + 0 \\ -0.333 - 0.667 + 0 \end{bmatrix} = \begin{bmatrix} 1.0 \\ -1.0 \end{bmatrix}$$

$$\delta^{(2)} = \begin{bmatrix} 1.0 \\ -1.0 \end{bmatrix} \odot \text{ReLU}' \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\delta^{(2)} = \begin{bmatrix} 1.000 \\ -1.000 \end{bmatrix} \checkmark$$

$$\Rightarrow \delta^{(1)} = (W_2^T \delta^{(2)}) \odot \text{ReLU}'(z^{(1)})$$

$$W_2^T \delta^{(2)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1.000 \\ -1.000 \end{bmatrix} = \begin{bmatrix} 1.000 \\ -1.000 \end{bmatrix}$$

$$\delta^{(1)} = \begin{bmatrix} 1.000 \\ -1.000 \end{bmatrix} \odot \text{ReLU}' \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$\delta^{(1)} = \begin{bmatrix} 1.000 \\ -1.000 \end{bmatrix} \checkmark$$

c) Compute all gradients and update:

$$W_3, b_3, W_2, b_2, W_1, b_1$$

$$\eta = 0.1$$

Layer 3 $\Rightarrow \nabla_{W_3} L = \delta^{(3)} (a^{(2)})^T$

$$= \begin{bmatrix} 0.333 \\ -0.667 \\ 0.333 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.333 & 0.333 \\ -0.667 & -0.667 \\ 0.333 & 0.333 \end{bmatrix} = \nabla_{W_3} L$$

$$\nabla_{b_3} L = \begin{bmatrix} 0.333 \\ -0.667 \\ 0.333 \end{bmatrix}$$

⇒ Update W_3 :

$$W_3(\text{new}) = W_3(\text{old}) - \eta \nabla_{W_3} L$$

$$= \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 0 \end{bmatrix} - 0.1 \begin{bmatrix} 0.333 & 0.333 \\ -0.667 & -0.667 \\ 0.333 & 0.333 \end{bmatrix}$$

$$W_3(\text{new}) = \begin{bmatrix} 0.967 & -1.033 \\ -0.933 & 1.067 \\ -0.033 & -0.033 \end{bmatrix}$$

Update b_3 ⇒ $b_3(\text{new}) = b_3(\text{old}) - \eta \nabla_{b_3} L$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 0.333 \\ -0.667 \\ 0.333 \end{bmatrix}$$

$$b_3(\text{new}) = \begin{bmatrix} -0.033 \\ 0.067 \\ -0.033 \end{bmatrix}$$

$$\Rightarrow \nabla_{w_2} L = \delta^{(2)} (a^{(1)})^T$$

$$= \begin{bmatrix} 1.000 \\ -1.000 \end{bmatrix} \begin{bmatrix} 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3.000 & 1.000 \\ -3.000 & -1.000 \end{bmatrix} = \nabla_{w_2} L$$

$$\nabla_{b_2} L = \delta^{(2)} = \begin{bmatrix} 1.000 \\ -1.000 \end{bmatrix}$$

Update w_2 :

$$w_{2(\text{new})} = w_{2(\text{old})} - \eta \nabla_{w_2} L$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 0.1 \begin{bmatrix} 3.000 & 1.000 \\ -3.000 & -1.000 \end{bmatrix}$$

$$w_{2(\text{new})} = \begin{bmatrix} 0.700 & -0.100 \\ 0.300 & 1.100 \end{bmatrix}$$

$$b_{2(\text{new})} = b_{2(\text{old})} - \eta \nabla_{b_2} L$$

$$= \begin{bmatrix} -2 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 1.000 \\ -1.000 \end{bmatrix}$$

$$= \begin{bmatrix} -2.100 \\ 0.100 \end{bmatrix} = b_{2(\text{new})}$$

$$\text{Layer 1} \Rightarrow \nabla_{W,L} = \delta^{(1)} x^T$$

$$= \begin{bmatrix} 1.000 \\ -1.000 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 2 \end{bmatrix}$$

$$\nabla_{W,L} = \begin{bmatrix} 1.000 & 0.000 & -1.000 & 2.000 \\ -1.000 & 0.000 & 1.000 & -2.000 \end{bmatrix}$$

$$\nabla_{b,L} = \delta^{(1)} = \begin{bmatrix} 1.000 \\ 1.000 \end{bmatrix}$$

$$\text{Updates: } W_{1(\text{new})} = W_{1(\text{old})} - \eta \nabla_{W,L}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \end{bmatrix} - 0.1 \begin{bmatrix} 1.000 & 0.000 & -1.000 & 2.000 \\ -1.000 & 0.000 & 1.000 & -2.000 \end{bmatrix}$$

$$W_{1(\text{new})} = \begin{bmatrix} 0.900 & 0.000 & 0.100 & 0.800 \\ 0.100 & 1.000 & -1.100 & 0.200 \end{bmatrix}$$

$$b_{1(\text{new})} = b_{1(\text{old})} - \eta \nabla_{b,L}$$

$$b_{1(\text{new})} \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 1.000 \\ 1.000 \end{bmatrix} = \begin{bmatrix} -0.100 \\ 0.100 \end{bmatrix}$$

d) Perform one new forward pass with the updated parameters and decide whether the probability of the correct class has increased.

Level 1

$$z^{(1)} = W_1 x + b_1 \quad \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}$$

$$W_1 x = \begin{bmatrix} 0.900 & 0.000 & 0.100 & 0.800 \\ 0.100 & 1.000 & -1.100 & 0.200 \end{bmatrix}$$

$$= \begin{bmatrix} 0.900 + 0 - 0.100 + 1.600 \\ 0.100 + 0 + 1.100 + 0.400 \end{bmatrix} = \begin{bmatrix} 2.400 \\ 1.600 \end{bmatrix}$$

$$W_1 x + b_1 = \begin{bmatrix} 2.400 \\ 1.600 \end{bmatrix} + \begin{bmatrix} -0.100 \\ 0.100 \end{bmatrix} = \begin{bmatrix} 2.300 \\ 1.700 \end{bmatrix} = z^{(1)} \quad \checkmark$$

$$a^{(1)} = \text{ReLU}(z^{(1)}) = \begin{bmatrix} 2.300 \\ 1.700 \end{bmatrix} \quad \checkmark$$

Layer 2

$$z^{(2)} = W_2 a^{(1)} + b_2$$

$$W_2 a^{(1)} = \begin{bmatrix} 0.700 & -0.100 \\ 0.300 & 1.100 \end{bmatrix} \begin{bmatrix} 2.300 \\ 1.700 \end{bmatrix} = \begin{bmatrix} 1.6100 - 0.170 \\ 0.690 + 1.870 \end{bmatrix}$$

$$W_2 a^{(1)} + b_2 = \begin{bmatrix} -0.660 \\ 2.660 \end{bmatrix} = z^{(2)}$$

$$a^{(2)} = \text{ReLU}(z^{(2)}) = \begin{bmatrix} 0.000 \\ 2.660 \end{bmatrix}$$

Layer 3

$$z^{(3)} = W_3 a^{(2)} + b_3$$

$$W_3 a^{(2)} + b_3 = \begin{bmatrix} 0.967 & -1.033 \\ -0.933 & 1.067 \\ -0.033 & -0.033 \end{bmatrix} \begin{bmatrix} 0.000 \\ 2.660 \end{bmatrix} + b_3$$

$$= \begin{bmatrix} 0.000 - 2.748 \\ 0.000 + 2.838 \\ 0.000 - 0.088 \end{bmatrix} + \begin{bmatrix} -0.033 \\ 0.067 \\ -0.033 \end{bmatrix}$$

$$z^{(3)} = \begin{bmatrix} -2.781 \\ 2.905 \\ -0.121 \end{bmatrix} \quad \checkmark$$

Soft Max: $e^{-2.781} \approx 0.062$, $e^{2.905} \approx 18.265$,

$e^{-0.121} \approx 0.886$ Sum: 19.213

$$\hat{y} = \begin{bmatrix} 0.062/19.213 \\ 18.265/19.213 \\ 0.886/19.213 \end{bmatrix} \approx \begin{bmatrix} 0.003 \\ 0.951 \\ 0.046 \end{bmatrix}$$

$\Rightarrow$ YES, the probability is much higher!